

Anu Prasanthi Pothula

Senior Software Engineer (React, Node, Java, AWS)

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Summary: Senior Software Engineer with 8 years of experience designing and building scalable web applications using React, Node.js, and cloud platforms. Experienced in developing end-to-end systems, including frontend architecture, backend APIs, and microservices using Node.js and Java (Spring Boot). Strong focus on performance optimization, system reliability, and clean architecture. Recently working on real-time AI integrations and building intelligent, user-centric applications.

Key Skills & Expertise:

- **Frontend Development:** React.js, React Native, Redux (Redux-Thunk, Redux-Saga, Redux Toolkit), React Router, MobX, Flux.js, Material-UI, Chakra UI, Bootstrap, TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, JSX, JSON
- **Styling & Design:** Styled-components, Emotion, BEM, CSS Grid, Flexbox, Media Queries, Responsive Design, CSS Modules, Figma, Sketch, Adobe XD
- **Performance Optimization:** Code Splitting, Lazy Loading, React Suspense, SSR (Next.js), SSG (Gatsby), React Profiler, Lighthouse, Redis, React Query, Caching Strategies
- **Testing & Debugging:** Jest, Mocha, Chai, Jasmine, Karma, Cypress, Protractor, React Testing Library, Chrome DevTools, Firebug
- **Backend Development:** Node.js, Java, Spring Boot, Spring MVC, Express.js, Microservices Architecture, RESTful APIs, SOAP, GraphQL, WebSockets
- **State Management:** Redux, Redux-Saga, Redux Thunk, React Context API, React Hooks
- **Databases:** MongoDB, MySQL, PostgreSQL, Oracle, MS SQL Server, Cassandra, SQL Query Optimization, Spring Data JPA, Hibernate
- **Cloud Technologies & DevOps:** AWS (Lambda, CloudFront, S3), Azure, Docker, Jenkins, GitHub Actions, CircleCI, Travis CI, Maven, Serverless Framework, AWS Lambda
- **Security:** JWT, OAuth 2.0, CORS Management, XSS Prevention, RBAC, CSP, Secure API Design, Input Sanitization
- **Version Control & Collaboration:** Git, GitHub, SVN, GitFlow, JIRA, Confluence, Trello
- **CI/CD & Automation:** Jenkins, GitHub Actions, CircleCI, Travis CI, Continuous Integration, Continuous Deployment
- **Agile Methodologies:** SCRUM, Sprint Planning, Backlog Grooming, Mentorship, Cross-Functional Collaboration
- **Mobile Development:** React Native, Progressive Web Apps (PWA), Mobile-First Design, Push Notifications, Offline Capabilities
- **Project Management & Leadership:** Team Lead, Cross-Functional Teams, Stakeholder Communication, Agile Coaching, Resource Allocation
- **Software Architecture & Design:** Microservices, SOA, Modular Design, Component-Based Architecture, Design Patterns (MVC, Singleton, Factory)
- **AI & Automation:** Hugging Face Transformers, Cartesia APIs, OpenAI APIs, AI voice agents, Cursor IDE, real-time LLM integration, AI-driven UI components

Core Competencies

- Leading the design and development of scalable, high-performance SPAs using React.js, Redux, and modern front-end libraries.
- Building and maintaining cloud-based solutions with AWS, ensuring continuous integration and delivery using Jenkins and Docker.
- Crafting seamless user experiences through advanced CSS, responsive design, and UI frameworks like Material-UI and Chakra UI.
- Handling API integrations and performance optimization with Redis, React Query, Lighthouse, and caching strategies.

Additional Skills:

- Expertise in version control and CI/CD pipelines for automation and seamless delivery.
- Experience with modern build tools like Webpack and exposure to Vite/ESBuild
- Testing and quality assurance with frameworks like Jest, Mocha, Chai, and Cypress to ensure clean

and maintainable code.

- Experienced in cloud services (AWS, Azure) and serverless architecture with AWS Lambda and CloudFront for optimized scaling.
- Strong knowledge of security best practices including JWT authentication, CORS, and XSS prevention to build secure applications

Certifications: React: <https://www.hackerrank.com/certificates/iframe/3dbe74721bed>

Education: Bachelor of Engineering (B.E.) in Electronics and Communication
Andhra University

2014 - 2018

Professional Experience

Senior Software Engineer

June 2025 - April 2026

CapitalOne, Canada

- Architected and implemented a scalable micro-frontend architecture, enabling independent deployments and reducing cross-team dependencies across enterprise applications
- Developed reusable and performant UI components using React (Hooks, Context API), ensuring consistency and maintainability across multiple modules
- Established robust testing practices using Playwright and Jest, achieving 90%+ coverage and improving release reliability
- Integrated Snowplow analytics to enable data-driven UX decisions and support A/B testing initiatives
- Optimized application performance through code splitting, lazy loading, and asset optimization, improving Core Web Vitals and overall user experience
- Built scalable backend services using Node.js and Express.js, designing RESTful APIs for high-performance web applications and enterprise platforms.
- Developed and maintained microservices architecture using Node.js, improving modularity, scalability, and system reliability across multiple applications.
- Familiar with modern state management tools like Zustand/Jotai and Context API
- Implemented CI/CD pipelines using GitLab CI, automating build, test, and deployment workflows
- Leveraged AI-assisted development tools such as Cursor IDE and Claude AI to accelerate coding, debugging, and system design workflows
- Used AI tools for code generation, refactoring, and performance optimization, improving development speed and code quality
- Strong understanding of React rendering lifecycle, reconciliation, and performance optimization techniques

Senior Software Engineer

Amiggi, Canada

July 2023 – June 2025 (Client: Vocal

Eats)

Project: Partner Portal

- Developed scalable, high-performance web applications using React, TypeScript, and modern frontend architectures, ensuring cross-browser compatibility and responsive design
- Implemented dynamic routing, lazy loading, and code splitting to enhance performance and reduce initial load times
- Translated complex Figma designs into pixel-perfect, production-ready UI component
- Built reusable component libraries using Material UI and modern styling approaches, improving development efficiency and UI consistency
- Optimized performance using SSR/SSG (Next.js, Gatsby), caching strategies (React Query, Redis), and Lighthouse-driven improvements, reducing latency and improving SEO
- Designed and implemented backend workflows in Node.js for handling user onboarding, notifications, payments, and third-party service integrations across scalable web applications.
- Designed secure authentication and authorization flows using JWT, OAuth2, and RBAC, ensuring application security and compliance
- Established comprehensive testing strategies using Jest, React Testing Library, and Cypress, integrated with CI/CD pipelines for reliable deployments
- Improved system performance by implementing caching and API optimization strategies, reducing

- response times by up to 40%
- Built and maintained CI/CD pipelines using Azure DevOps, improving deployment efficiency and release reliability
- Integrated third-party services (Stripe, Firebase, Google Maps) to enable real-time features, payments, and geolocation capabilities
- Deployed and maintained applications on AWS (S3, CloudFront, Lambda) and modern hosting platforms, ensuring scalability and high availability
- Contributed to scalable system design using modular architecture and design patterns, improving long-term maintainability

Technical Lead

Deloitte (Client: Axis Bank)

Nov 2022 – June 2023

Project: Axis Mutual Funds

- Led frontend architecture and delivery of high-performance web applications using React and modern JavaScript, ensuring scalability and timely execution
- Designed and implemented modular, maintainable frontend systems with efficient state management (Redux Toolkit) and dynamic routing
- Collaborated cross-functionally with product, design, and backend teams to deliver seamless, user-centric features aligned with business goals
- Improved application performance through code splitting, lazy loading, and optimization techniques, enhancing load times and user experience
- Contributed to backend API design and integration using Node.js and Express, supporting real-time features with WebSockets
- Optimized critical backend endpoints, improving response times by up to 20% and enhancing overall system performance
- Implemented secure authentication and authorization mechanisms using JWT, RBAC, and best practices for data protection
- Established CI/CD pipelines and testing practices, enabling reliable deployments and improving development workflows
- Mentored developers, conducted code reviews, and drove engineering best practices to maintain high code quality and team productivity
- Acted as a Subject Matter Expert (SME) for frontend architecture, performance optimization, and best practices
- Implemented accessible UI using ARIA roles, semantic HTML, and WCAG standards

Software Engineer

Mphasis Limited,

May 2021 – Oct 2022

Project-1: Edge Price Forbes

- Developed Edge Price Forbes Online using React.js for dynamic front-end development and Node.js for back-end services, with Redux for state management and React Router for efficient navigation.
- Designed and documented the project using Confluence, creating mockups and maintaining clear communication with team members.
- Built responsive user interfaces with Material-UI, incorporating CSS Flexbox, Grid, and media queries for mobile-friendly layouts, ensuring cross-browser compatibility.
- Developed reusable React components for forms, lists, and tables, leveraging React Hooks for state and effect management to create modular and scalable UI elements.
- Ensured secure authentication through Role-Based Access Control (RBAC) and JWT for token-based user authentication.
- Integrated third-party APIs using Axios, connecting the front-end with back-end services for dynamic data fetching.
- Created interactive dashboards with libraries such as Chart.js and D3.js, displaying real-time data visualizations.
- Optimized app performance through techniques like Code Splitting, Lazy Loading, and React Profiler, reducing load times and improving user experience.
- Collaborated using Agile methodologies, participating in sprint planning, backlog grooming, and code reviews with tools like JIRA and Git for version control.

- Deployed the application on AWS using Docker for containerization, ensuring scalability and portability in cloud environments.

Project-2: RBS (Royal Bank of Scotland)

- Developed responsive web pages using HTML5, CSS3, Material-UI, and React.js, ensuring cross-browser compatibility and a smooth user interface.
- Created reusable React components and managed state with Redux, implementing unidirectional data flow and using lifecycle methods, props, and state for efficient data transfer and UI updates.
- Utilized GitHub for version control, managing code branches, and deploying the application.
- Implemented Ajax calls to handle dynamic data fetching via JSON and integrated server-side content using Contentful.

Software Engineer

Adecco (Client: WayCool Foods & Products Pvt Ltd)

Nov 2020 – April 2021

Project: My-Price

- Used React.js with Redux Flux for logical programming and Material-UI for beautification.
- Managed the state of the application using Redux, ensuring that user sessions, pricing data, and other essential application states were consistently handled across the app.
- Employed Object-Oriented Programming principles to create reusable UI components and enhance modularity across web applications.
- Employed JavaScript (ES6+) to implement various features like form validation, user input processing, and API call handling, ensuring a smooth user experience.
- Implemented login, validations, roles, latitude, and longitude for location control.
- Used Axios for HTTP/HTTPS requests and attached Bearer token for secure usage of API.
- Built RESTful APIs using Spring Boot and Spring MVC to handle user authentication, pricing data, and transaction management for the My-Price platform
- Integrated databases like MongoDB and MySQL to ensure efficient data storage and retrieval for applications.
- Facilitated communication between applications by integrating SOAP and RESTful web services, ensuring efficient data exchange and smooth operations.
- Used Spring Data JPA with Hibernate for efficient handling of MongoDB data, optimizing performance and scalability.

Software Developer

Nipun Net Solutions

June 2018 – Oct 2020

Project: Asset Management

- Developed responsive and modular front-end interfaces using React.js, HTML5, CSS3, and ES6+, ensuring real-time updates and seamless user interactions across devices.
- Integrated Axios to manage secure HTTP requests and synchronize data between frontend and backend services.
- Engineered backend services with Java 8 and Spring Boot, building scalable microservices and RESTful CRUD APIs for asset management and administrative workflows.
- Utilized Spring Data JPA and Hibernate for ORM, enabling efficient data operations with MySQL, and implemented Spring Security for role-based authentication and access control.
- Designed and optimized data handling using Java Collections (List, Map, Set) to support complex asset-related logic.
- Ensured reliable communication between frontend and backend by deploying Spring MVC-based endpoints.
- Wrote unit tests with JUnit to maintain code reliability and quality and leveraged Jenkins and Maven to automate CI/CD pipelines and build processes.

PERSONAL PROJECTS

- Built a voice-interactive landing page using React, TypeScript, and Tailwind CSS with real-time AI API integrations for speech-based interactions
- Implemented end-to-end voice interaction flow using speech-to-text and text-to-speech APIs, handling async events and managing conversational state